Multi-Rail Isolated Power Supply Monitoring System

Capstone Design Documentation

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Contents

1 Introduction **1**

1.1 Project Overview 1

1.2 Project Motivation 2

1.3 System Architecture 2

1.4 Design Objectives 2

1.5 Project Scope 3

1.6 Report Organization 4

2 Problem Background and Definition **5**

2.1 Purpose and Scope 5

2.2 Problem Definition 5

2.3 Proposed Solution 6

3 Technology Solution **7**

4 Chosen Conceptual Design **9**

4.0.1 Functional Block Diagram 10

5 Potential Social, Ethical, Legal, and Environmental Impacts of the Project **11**

5.1 Societal Impact 11

5.2 Global Impact 11

5.3 Legal and Regulatory Impact 12

5.4 Ethical Impact 12

5.5 Environmental Impact 13

5.6 Relationship to the Original Requirements 13

6 Design Methodology **15**

7 Simulation Results **17**

8 Evaluation of Solution's Efficacy **19**

9 Improvements and Optimizations to Solution **21**

10 Conclusions **23**

10.0.1 Did the Project Meet the Requirements of the Scope? 23

10.0.2 Was the Project Completed on Time? 23

10.0.3 Was the Project Completed Under Budget? 24

10.0.4 Did the Quality and Performance Meet Stakeholder Expectations? . . 24

10.0.5 Overall Project Assessment 25

11 References **27**

Chapter 1

Introduction

1.1 Project Overview

The Multi-Rail Isolated Power Supply Monitoring System is a hardware and software system developed to provide centralized monitoring of multiple electrically isolated power-supply rails. The system measures both output voltage and load current, provides configurable alarm limits, displays operating status and measurement data via a local display or ethernet for integration with an EPICS-based control system.

The completed system monitors six independent power-supply rails:

- +5VA
- +5VB
- +5V
- -5V
- +15V
- -15V

Because the monitored supplies may be electrically isolated from one another, the monitoring system must preserve that isolation while still allowing all six channels to communicate with a common controller. Each supply rail therefore uses an independent voltage/current monitoring channel and an isolated digital communication path.

The primary controller is an Arduino GIGA R1 WiFi. The controller acquires measurement data from the individual monitor channels, converts the data to engineering units, applies the appropriate rail polarity, evaluates configured alarm limits, and presents the resulting information on a GIGA Display Shield. An Ethernet interface provides network communication between the embedded controller and an EPICS Input/Output Controller (IOC). The EPICS IOC then makes the measurements, alarms, and system status available to a CS-Studio Phoebus operator interface.

1.2 Project Motivation

Power-supply systems containing multiple isolated voltage rails present a monitoring challenge because each circuit associated with one supply can't share a direct electrical reference with another supply. Connecting conventional measurement or communication circuits across these references can break isolation between the rails.

A centralized monitor is desirable because it allows the user to observe all supply voltages and currents from a single interface, identify abnormal operating conditions, detect communication or sensor faults, and integrate the power-supply status into a larger control system.

The design developed for this project addresses these requirements by performing voltage and high-side current measurement locally on each monitored rail and transferring the resulting information across electrically isolated digital interfaces. The controller receives measurement data from all six rails without requiring their power references to be directly connected together.

1.3 System Architecture

The monitoring system is divided into three primary functional areas: the power-supply monitoring hardware, the microcontroller, and the supervisory control interface.

The power supply monitoring hardware contains the current-sense resistors, LTC2945 PMICs, and digital isolation circuitry. Each LTC2945 measures the supply voltage and the differential voltage developed across a high-side current-sense resistor. The resulting digital measurement information is sent over the I^2C bus while maintaining the required electrical isolation.

The Arduino GIGA R1 WiFi provides the mcu processing functions. Its firmware initializes the monitoring channels, acquires voltage and current measurements, detects communication errors, converts raw measurements to engineering units, applies channel-specific polarity and scaling, evaluates alarm limits, and updates the local display. Configuration information, including network parameters and alarm limits, is retained in persistent storage so that required settings can survive a controller reset or power cycle.

Network communication is provided through an Ethernet interface using the PSC protocol. The EPICS IOC communicates with the embedded controller using PSCDriver and converts the transmitted values into EPICS process variables. CS-Studio Phoebus provides the remote operator interface for viewing measurements, alarm conditions, connection status, and system state.

1.4 Design Objectives

The major design objectives for the Multi-Rail Isolated Power Supply Monitoring System are to:

1. Measure the voltage of six independent power-supply rails.
2. Measure the load current of each rail using high-side current sensing.
3. Preserve electrical isolation between independently referenced power-supply rails and the common controller.
4. Convert acquired data into meaningful engineering units and correctly represent both positive and negative voltage rails.
5. Detect I^2C communication failures and prevent invalid sensor data from being presented as valid measurements.
6. Compare voltage and current measurements against configurable upper and lower alarm limits.
7. Provide a local graphical display of voltage, current, alarm status, communication status, and network state.
8. Retain configuration and alarm settings through controller resets and power cycles.
9. Provide Ethernet communication for remote monitoring and configuration.
10. Integrate the monitoring system with an EPICS IOC and CS-Studio Phoebus operator interface.
11. Maintain the hardware, firmware, and control-system software under revision control so that changes made during design, implementation, and testing remain traceable.

1.5 Project Scope

The scope of the project includes the design and implementation of the sensor interface hardware, Arduino GIGA controller hardware, embedded firmware, Ethernet communication interface, EPICS IOC, and operator-interface software. The hardware portion includes the six voltage/current monitoring channels, current-sense resistors, digital isolation circuitry, controller interface, Ethernet circuitry, and supporting power-supply circuitry.

The software portion includes the embedded acquisition and alarm-processing firmware, persistent configuration storage, local display interface, PSC network communication, EPICS IOC configuration and database records, and the CS-Studio Phoebus operator interface.

Design files and software associated with the project are maintained in the project Git repository:

`https://github.com/capotosto/PS_Monitor`

The repository provides revision control for the hardware design, firmware, EPICS IOC files, documentation, and other project artifacts generated during development.

1.6 Report Organization

This report documents the development of the monitoring system from preliminary design through implementation and testing. The preliminary-design chapter defines the major functional blocks, presents the system architecture, documents supporting engineering calculations, includes the hardware schematics and bills of materials, and describes the primary firmware and control-system software flows.

Subsequent chapters document prototyping and development activities, the EPICS IOC implementation, and the CS-Studio Phoebus operator interface. Supporting project-management and design documents developed during the earlier phases of the capstone project are retained as part of the overall project documentation and provide traceability between the original requirements, design decisions, implementation, and final system.

Chapter 2

Problem Background and Definition

2.1 Purpose and Scope

This chapter documents the hardware and software design of the Multi-Rail Isolated Power Supply Monitoring System. The design monitors six power-supply rails, measures rail voltage and load current, preserves electrical isolation between floating supplies and the controller, evaluates measurements against user-configurable alarm limits, presents local status on the controller display, and publishes measurements and status to an EPICS control-system interface.

This design is intended to establish the major analog and digital functional blocks, document supporting engineering calculations, define the hardware schematic baseline, and provide software flowcharts detailed enough to support implementation.

2.2 Problem Definition

The project addresses the need to monitor several low-voltage power-supply rails that do not necessarily share a common reference. A conventional monitoring circuit that directly connects the grounds together would defeat the electrical isolation of the supplies and could create unwanted return paths. However, the user needs a single location from which to observe rail voltage, rail current, alarm state, communication status, and the overall condition of the power system.

The required monitoring system must therefore make accurate measurements on six separate rails while preserving isolation between the monitored domains. It must support both positive and negative rails, detect sensor or communication faults, provide configurable voltage and current limits, retain configuration information after reset, present useful local status, and make the same data available to the facility control system over Ethernet. The monitored rails in the present design are +5 V A, +5 V B, +/-5 V, and +/-15 V.

2.3 Proposed Solution

The proposed solution assigns one LTC2945 voltage/current monitor to each power rail. Load current is converted to a small differential voltage by a precision high-side shunt resistor and the LTC2945 digitizes both the shunt voltage and the rail voltage. Each sensor remains referenced to its own monitored supply domain. The I^2C communication path is then isolated so that digital measurement data can be transferred to a common Arduino GIGA R1 WiFi controller without directly bonding the isolated supply returns together.

The Arduino GIGA performs channel acquisition, engineering-unit conversion, negative-rail sign handling, alarm evaluation, configuration storage, display updates, and network communication. A local GIGA Display Shield provides immediate status at the hardware. Ethernet transports measurement and status data using the PSC protocol to an EPICS IOC, where PSCDriver and EPICS records make the values available to a CS-Studio Phoebus operator interface. This architecture separates the rail-side analog measurement functions from the controller and supervisory-control functions while maintaining a single user interface for the complete system.

Chapter 3

Technology Solution

Chapter 4

Chosen Conceptual Design

The major system functions are summarized below.

1. **Power-Supply Interface:** Connect to the six monitored rails and route each rail through the associated voltage/current monitoring channel.
2. **Current Measurement:** Convert load current to a differential voltage across a precision shunt resistor and digitize it with the LTC2945.
3. **Voltage Measurement:** Measure the monitored rail voltage using the LTC2945 supply-voltage ADC channel.
4. **Electrical Isolation:** Maintain galvanic isolation between each floating monitored supply and the common controller-side I^2C bus.
5. **Digital Acquisition:** Poll each monitor, identify communication faults, apply channel-specific sign/scaling information, and calculate engineering-unit values.
6. **Alarm Processing:** Compare voltage and current values against configured upper and lower limits and generate channel alarm states.
7. **Local User Interface:** Present all rail measurements, alarms, network state, and communication status on the Arduino GIGA Display.
8. **Network/EPICS Interface:** Publish measurements, alarm limits, alarm state, and connection status over Ethernet for the EPICS IOC and Phoebus interface.

4.0.1 Functional Block Diagram

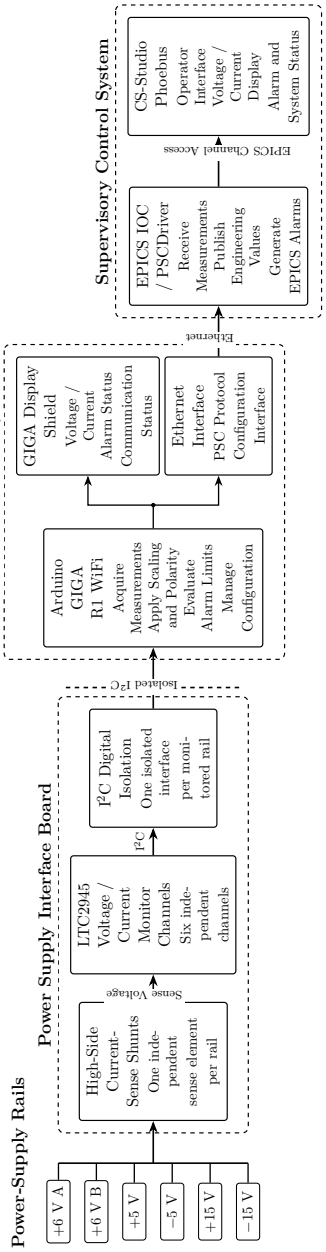


Figure 4.1: Functional block diagram of the multi-rail isolated power supply monitoring system. Six independent power-supply rails are measured using high-side current sensing and LTC2945 voltage/current monitoring channels. Digital isolation preserves electrical independence between the monitored supplies and the common controller. The Arduino GIGA processes the measurements, provides a local operator display, and transmits measurement and status information over Ethernet to the EPICS IOC and CS-Studio Phoebus operator interface.

Chapter 5

Potential Social, Ethical, Legal, and Environmental Impacts of the Project

5.1 Societal Impact

The principal societal impact of the monitoring system is the improved data availability. The six isolated power supplies that would otherwise have to be monitored individually can now be measured from one local display and from the EPICS control system. Abnormal voltage, current, or communication conditions can be detected faster. In a laboratory or industrial setting, this can reduce troubleshooting time and help to avoid operation of the equipment with a power supply failure.

There is also a obligation associated with presenting measurements to an operator. Erroneous data can be more problematic than unavailable data if an operator believes that a failed measurement is a good one. For this reason, communication failure detection was included in the design. The firmware distinguishes an failed or offline sensor from a legitimate zero measurement, and presents the communication state together with the measured values. This was an important design decision since the monitor is meant to present information which an operator can rely on for making maintenance and operational decisions.

5.2 Global Impact

The design is dependent on a global electronics supply chain. ICs, passive components, PCB fabrication, and commercial Arduino modules can be manufactured in several different countries even when they are supplied through United States distributors. This gives access to a wide range of components and lower manufacturing costs, but at the same time creates risks associated with component availability, obsolescence, shipping disruptions, and supplier dependence.

The design mitigates some of these risks through the use of standard component packages, conventional PCB assembly, repetitive circuit blocks, and standard interfaces such as I^2C

and Ethernet. makes qualification of replacement components or alternate manufacturing sources much easier than it would be with highly proprietary hardware.

Global sourcing creates an ethical obligation that is not completely covered by component price and technical performance. For a production version of the monitor, the selection of a supplier should also include consideration of responsible labor practices, material sourcing, regulatory compliance, and traceability of the supplier. These considerations considerations would become increasingly important for large scale production of the design.

5.3 Legal and Regulatory Impact

This project is an engineering prototype and can't be considered to have any system-level regulatory approval solely because individual components carry approvals or compliance markings. If the monitor were converted into a commercial product, the required regulatory evaluation would depend on the installation, enclosure, power source, network interfaces, and markets in which it would be sold.

For example, no attention was paid to RoHS compliance, as the boards are intentionally assembled with leaded solder, so leaded HASL or ENIG were both acceptable ordering options. Use of the monitor with leaded solder however, would be an issue for many markets with RoHS mandates.

EMC, product-safety, and radio-frequency requirements would also have to be identified for the final installation before the production product could be offered for sale. EMC lab compliance testing can be extremely expensive.

The distinction between component compliance and system compliance is therefore important. The design can take advantage of the recognized components and accepted engineering practice without implying any approval not obtained for the finished assembly. This can most easily be related to the idea of "UL Listed" and "UL Registered" (UL vs UR labels) - UL listed components are certified on their own standing, but UR components need to be certified in the installation, and are only a recognized component of a system.

5.4 Ethical Impact

The first ethical obligation associated with the monitor is the accurate and unambiguous presentation of the measured information. The system should not hide sensor failures or substitute erroneous data with apparently valid measurements. The implemented communication failure detection, alarm indication, and separate sensor failure reporting were included partly for this reason.

The next consideration is network security. The current HTTP and PSC interfaces are designed for use in a controlled local network. They do not have security features necessary to be connected to an untrusted network. If the design were deployed in an unprotected network, additional requirements for authentication, access control, encryption, network seg-

mentation, and configuration management should be dfined. Any possibility of unauthorized modification of alarm limits or the network configuration would undermine the reliability of the monitoring system, even if the measurement hardware would continue operating properly.

Ethical considerations also also relevant to procurement. Cost and availability are important engineering considerations, but they should not be the only criteria for selecting a production supplier. Labor practices, material sourcing, counterfeit parts risk, and the ability to provide traceable parts are also appropriate considerations for a product moving from prototype quantities into production.

5.5 Environmental Impact

The main environmental impacts of the monitor come from PCB fabrication, semiconductor manufacturing, raw-materials, transportation, electricity consumption, and eventual electronic waste. It's impossible to make a design with zero environmental impact, but some aspects of this design reduce the quantity of hardware that would need to be disposed of during the useful service life of the product.

The Sensor Board, Arduino Shield, Arduino GIGA, display, and other major assemblies are separable. Therefore, a failed subsystem can therefore be repaired or replaced independent of the complete monitoring system. Standard component packages and commercially available modules were preferred where possible, and this increases the probability of replacement of an individual component. Alarm limits and network configuration are stored in software which allows changing the configuration without producing a new PCB.

For production quantities, RoHS-compliant components should be used where possible. The end-of-life plan should include recovery or recycling of the electronic assemblies. These considerations align with RoHS and WEEE requirements to encourage collection, reuse, recovery, and recycling of electronic equipment.

5.6 Relationship to the Original Requirements

Several of the societal and ethical considerations described above were indirectly represented in the original requirements definition. Electrical isolation was explicitly required to allow the monitoring of one rail would not defeat the isolation of the other. The communication fault detection was required to ensure erroneous data would not be interpreted as a valid electrical measurement. User-configurable alarm limits, local status indication, EPICS monitoring, and persistent configuration storage were also included as functional requirements. These requirements increase the reliability, maintainability, and the quality of the information presented to the operator.

Environmental considerations were introduced later in the design through serviceability, component reuse, and the use of common parts. Regulatory certification, ethical supplier selection, and cybersecurity requirements were not among major items in the original re-

quirements definition. If the prototype were developed into a production product, these aspects should be transformed into explicit requirements with acceptance criteria.

Chapter 6

Design Methodology

Chapter 7

Simulation Results

Simulations were performed using LTSpice in a DC Operating Point simulation to verify the calculation for the full-load shunt voltage for each of the voltages. Only the positive polarity was tested in the simulation, as negative polarity will naturally not matter to a a passive element such as the resistor.
The data collected in this simulation matches the data found in Appendix A section **??**, table **??**

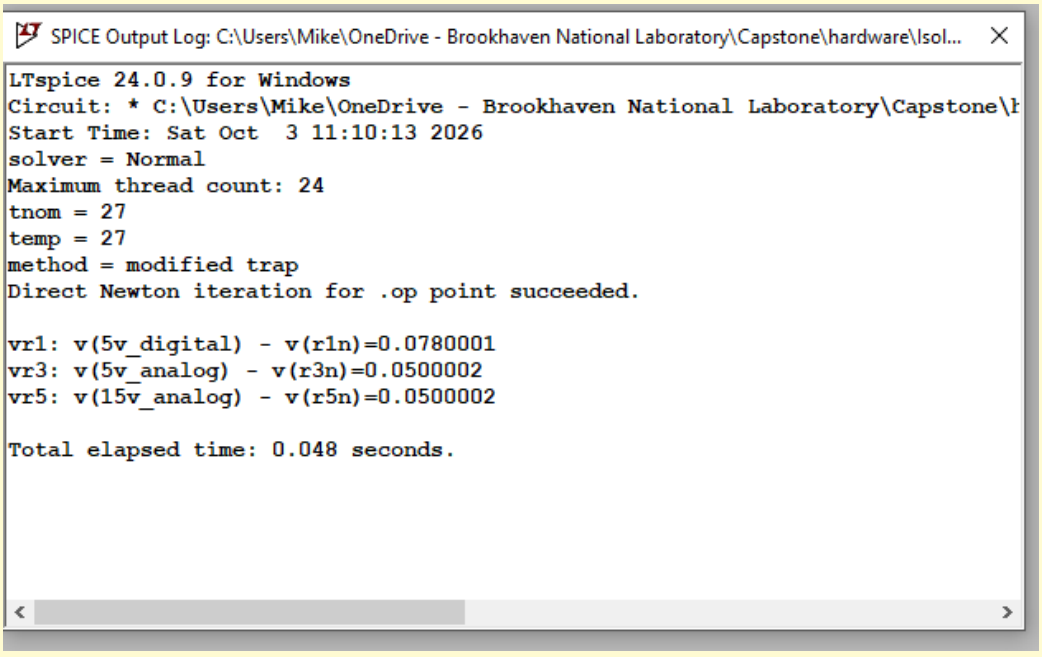


Figure 7.1: Simulation verifying current shunt voltage drops, data log

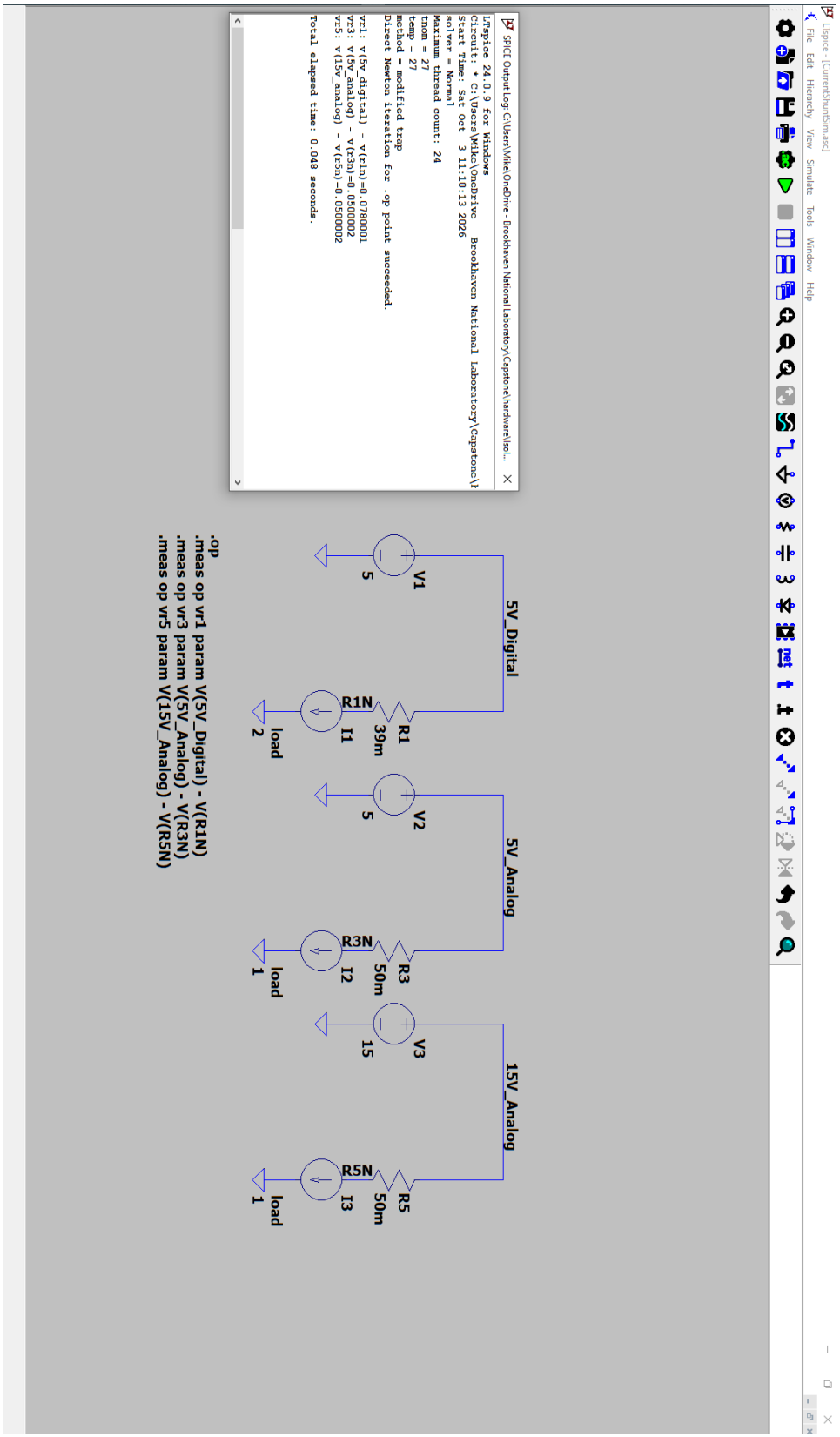


Figure 7.2: Simulation verifying current shunt voltage drops, full simulation screenshot

Chapter 8

Final Design

Chapter 9

Solution Implementation

9.1 Testing the LTC2945

The LTC2945 testing was carried out using a Linear Tech eval board, DEMO CIRCUIT 1697A. Connections and modifications were made as follows:

- VIN: tied to Arduino +5V
- GND: tied to Arduino GND
- ADIN: Tied to GND
- SCL: Tied to Arduino Pin D8 I2C2 SCL
- SDAI/SDAO (JP2 set to TIE): Arduino Pin D9 I2C2 SDA
- VPU: Tied to Arduino +3v3
- VOUT: Float or tied to dummy load (1kΩ to 10kΩ)
- 20mΩ shunt replaced with 17Ω shunt for testing with ∼5kΩ load
- Installed 4.7kΩ pull up on I2C lines via R4, R5 footprints on EVB

9.1.1 Calculating Shunt Value for Test

Using the 5V from the arduino as a source, and a load ranging 1k to 10k ohms, we calculate:

$$R_{L,min} = 1k\Omega$$

$$V_{shunt} = V_S \frac{R_S}{R_L + R_S}$$

$$R_S = \frac{V_{Shunt} R_L}{V_s - V_{Shunt}}$$

$$V_{shunt} = 0.1024V$$

$$R_{S,fullscale} = \frac{(0.1024V)(1k\Omega)}{5V - 0.1024V} = 20.91\Omega$$

Adding some headroom, say, 15%, and going to an E96 resistor value...

$$R_S = 0.85(20.91) = 17.8\Omega$$

Verifying for the high end of the range:

$$I_{min} = \frac{5V}{10k\Omega + 17.8\Omega} = 499.1\mu A$$

$$V_{shunt,min} = (499.1\mu A)(17.8\Omega) = 8.88mV$$

The shunt will range from 8.88mV to 87.44mV, within the ADC's usable range of 0V to 102.4mV.

9.1.2 Pictures of Evaluation Board, Display, and Phoebus Panels

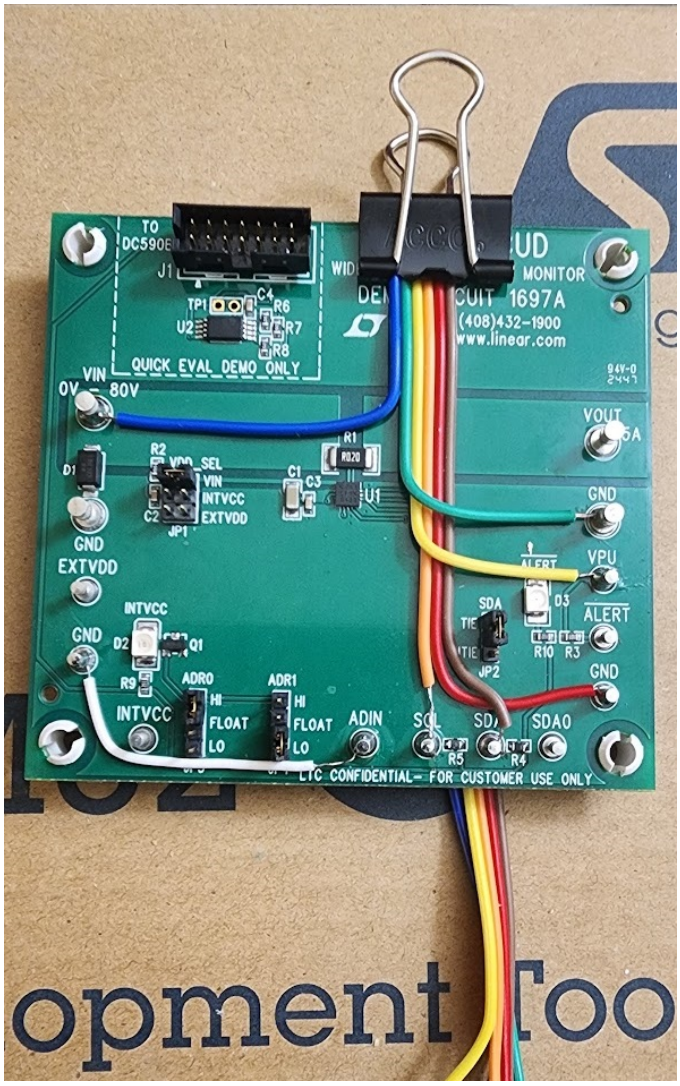


Figure 9.1: DEMO CIRCUIT 1697A Eval Board with Original Shunt, fly leads to connect to Arduino.



Figure 9.2: DEMO CIRCUIT 1697A Eval Board with 17Ω Shunt, fly leads to connect to Arduino, and 4.7kΩ test load.

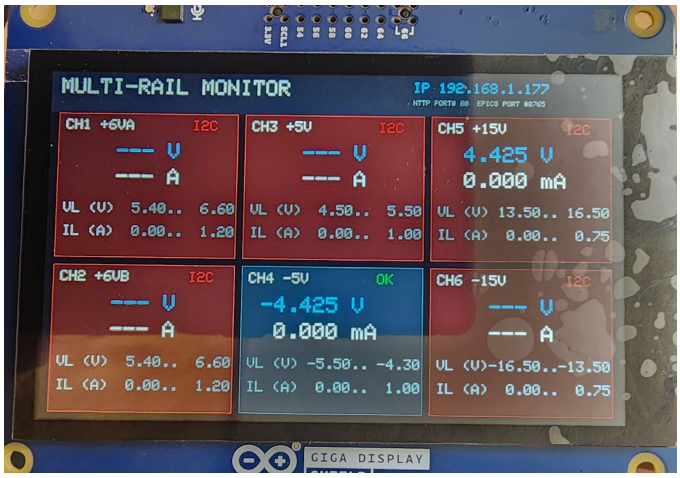


Figure 9.3: Arduino Giga with GIGA DISPLAY SHIELD showing dummy -5V rail being monitored, without alarm

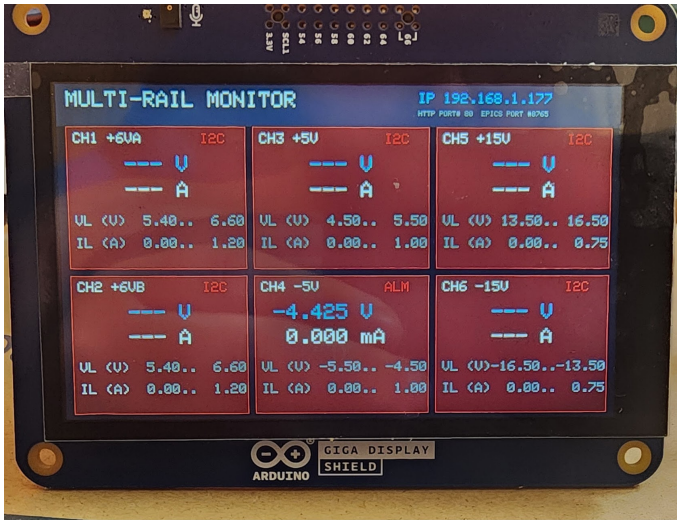


Figure 9.4: Arduino Giga with GIGA DISPLAY SHIELD showing dummy -5V rail being monitored, with UVL alarm, other channels I2C ALARM.

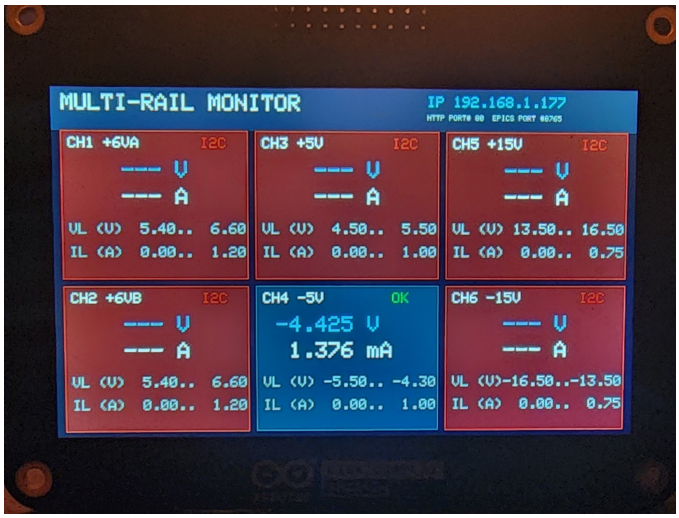


Figure 9.5: Arduino Giga with GIGA DISPLAY SHIELD showing dummy -5V rail being monitored, without alarm, 3.3kΩ Load



Figure 9.6: Arduino Giga with GIGA DISPLAY SHIELD showing dummy -5V rail being monitored, without alarm, 3.3kΩ Load



Figure 9.7: Arduino Giga with GIGA DISPLAY SHIELD connected to an LTC2945 eval board

Chapter 10

Evaluation of Solution's Efficacy

Chapter 11

Improvements and Optimizations to Solution

Despite the fact that the completed system satisfied the acceptance criteria, some improvements are recommended for the future PCB revision.

The most important hardware improvement is the incorporation of the SDA polarity fix directly in the PCB design. The prototype required the SN74LVC2G04 inverter to correct the SDA signal polarity after the originally installed LTC2945 device variant turned out to be incompatible with the intended isolation interface. The future revision should use the correct LTC2945 variant throughout the design or incorporate the required inversion directly in the schematic and PCB.

All required I^2C pull-up resistors and defined logic states should also be incorporated directly in the next PCB revision. It will eliminate the prototype modification and improve the repeatability of assembly and startup.

The firmware initialization can also be made more tolerant of arbitrary power-up sequencing. During the development, the communication could be failed when the controller tried to initialize the sensor before the corresponding isolated rail circuitry was powered. The future firmware revision could periodically retry the unavailable sensors so that the communication is established automatically when the monitored rail is powered.

The per-channel measurement calibration is another potential improvement. The present system calculates the current using the configured current sense resistance. The measurement of the actual resistance or the determination of the calibration factor against the external reference will allow the individual gain correction to be stored for each channel. It will help to reduce the systematic errors caused by the shunt tolerance and allow tighter acceptance limits if the improved measurement accuracy is required.

The additional environmental characterization is also useful if the design is going to be developed beyond the capstone prototype. The testing at multiple ambient temperatures can be used to determine whether the temperature affects the measurement accuracy. EMC testing can determine the susceptibility to and generation of the electromagnetic interference.

The long duration testing can be used to detect the intermittent communication, thermal, network or firmware problems that might not occur during the shorter acceptance test.

The automated test system should be retained for the future revisions. Since the raw measurements are saved separately from the report generation, the future hardware or firmware revisions can be tested using the same acceptance procedure and direct comparison between the revisions is possible.

These recommendations represent the refinements in reliability, manufacturability, calibration and qualification and not the changes to the basic architecture. The successful final test proves that the overall system concept is suitable for the further development.

Chapter 12

Conclusions

12.0.1 Did the Project Meet the Requirements of the Scope?

The project fulfilled the major requirements of the original project scope.

The completed system provides six independent voltage and current monitoring channels corresponding to +5VA, +5VB, +5 V, −5 V, +15 V, and −15 V. The high side current sensing is used for each channel while the electrical isolation is preserved between independently referenced monitored supplies and the common controller.

The completed hardware includes the sensor interface, current sense resistors, isolation circuitry, Arduino GIGA controller interface, local display, controller power circuitry and Ethernet interface. The software implementation includes the embedded measurement acquisition, conversion to engineering units, configurable limits, alarm processing, persistent configuration storage, local graphical display, network communication, the EPICS IOC, process variable database records and the Phoebus operator interface.

The final system testing confirmed the correct operation of all six measurement channels, the correct positive and negative voltage representation, programmable limit operation, persistent settings, local display operation, Ethernet communication, EPICS monitoring, communication recovery, repeatability, and end-to-end operation.

The hardware, firmware, IOC files, operator interface files, documentation and other project artifacts were also maintained under revision control as required by the project scope.

Based on the completed implementation and the final system test results, the capstone project is therefore considered to have fulfilled its technical scope.

12.0.2 Was the Project Completed on Time?

The project was completed within the planned capstone schedule.

The original project plan distributed the development across the eight week ET410 planning phase and the following eight week ET450 implementation phase, with the completed system

expected to be ready for demonstration by Week 7 of ET450. The final automated system test was completed on October 18, 2026, giving the completed and verified system before the planned final demonstration period.

The project experienced the schedule risk during the hardware bring-up. Troubleshooting of the isolated I^2C interface required additional effort after the SDA polarity problem and other communication problems were discovered. The startup sequencing and Ethernet integration also required the troubleshooting that was not anticipated in the initial plan.

These problems consumed the schedule margin, but they were fixed without preventing the completion of the prototype and final system test within the capstone schedule. The project is therefore considered to have fulfilled the schedule requirement.

12.0.3 Was the Project Completed Under Budget?

The project was not completed within the original preliminary hardware budget.

The initial project proposal assumed the material and contingency budget of $660. This estimate was based on the incomplete information about the detailed component selection, custom PCB layout and final vendor pricing.

After completion of the detailed design and Bill of Materials, the estimated one system hardware purchase cost became $1,019.03. It includes $242.36 in the Sensor Board components, $473.57 for the minimum Sensor Board PCB order, $180.90 in the Arduino Shield components, and $122.19 for the minimum Arduino Shield PCB order.

Compared to the original $660 estimate, the final documented hardware purchase estimate is $359.03 higher than the preliminary budget. The additional cost is due to a better definition of the component as well as the minimum order cost related to the fabrication of the custom PCBs.

Thus, the cost estimate was good to use as a preliminary cost estimate value but did not provide enough cost to build a complete prototype of a custom PCB. The project should be considered above the original hardware budget.

This cost comparison concerns only the documented prototype hardware cost. Laboratory test equipment, software, labor, shipping, taxes, enclosure hardware, cabling and any other items not included into the project cost analysis are not included into this cost comparison.

12.0.4 Did the Quality and Performance Meet Stakeholder Expectations?

The quality and performance of the completed prototype met the expectations set for the capstone project.

The main project stakeholders are the project developer, the course instructor and the potential technical users of the system. The best evidence of system quality is the completed acceptance test. All twelve expected system tests were completed successfully, all six mon-

itored rails passed the voltage and current check, and the complete system demonstration was successful without any system failures.

The prototype showed the capability to collect isolated voltage and current measurements, display the data locally, compare the data with configurable limits, keep the configuration after the restart, send the data via Ethernet, support the EPICS integration, present the data using Phoebus and to restore the operation after the network connection interruption.

The project provided the possibility to develop the repeatable automated test procedure instead of using the subjective one. The raw data were saved and used for the creation of a test report including the measurement data, pass/fail result and the test evidence. It improves the confidence in the final result and allows assessing future revisions of the hardware.

The prototype still includes some features which should be improved in the future production revision of the design, such as SDA inverter modification, modified implementation of the I^2C pull-up, correct startup sequence and the possible implementation of the per-channel calibration. They do not prevent the prototype from meeting the acceptance criteria, but may influence the manufacturability and further improvements of the design.

Based on the final test results, the completed project can be stated to meet the stakeholder expectations in terms of functionality and performance.

12.0.5 Overall Project Assessment

The Multi-Rail Isolated Power Supply Monitoring System successfully passed the full project cycle including the project definition and preliminary design, custom hardware development, firmware implementation, network and EPICS integration, board bring-up, troubleshooting and system verification.

The biggest challenges of the project were related to the hardware integration instead of development of the individual functions of the system. The troubleshooting of the isolated I^2C interface showed the necessity to check the component version, polarity, pull-up requirement, signal states and startup behavior in the physical implementation. These issues were found by the measurements, fixed and retested in the final system tests.

The completed prototype met the defined project scope, completed in the required time and met the final technical acceptance criteria. Although the project cost exceeds the original preliminary hardware budget, the detailed cost estimation provides more realistic basis for the estimation of the cost of prototype and future production revision.

Finally, the successful completion of the system test showed that the hardware, embedded firmware, local user interface, Ethernet communication, EPICS IOC, Phoebus interface and the automated test software work as the integrated system. Thus, the project successfully reached the main goal of development and demonstration of the functioning six-channel isolated power supply monitoring system.

Chapter 13

References